

GaN RF Industry:

Landscape and Future Evolution

Zhen ZONG
Analyst
zong@yole.fr

1 — GaN's added values

Why do we need GaN in RF industry?

2 — GaN RF players and products

Who are selling and in which market?

3 — GaN RF market size and evolution

What's the industry landscape?

4 — Wireless infrastructure and future development

The most important market segment

5 — Comparison GaN-on-SiC & GaN-on-Silicon

Which technology has more potential?

6 — Conclusion

Material	Breakdown electric field (MV/cm)	Thermal conductivity (W/cm/K)	Electron density ($10^{13}/\text{cm}^2$)	Mobility ($\text{cm}^2/\text{V/s}$)	Saturation velocity (cm/s)
Si	0.6	1.5	0.3	1300	1×10^7
GaAs	0.5	0.5	0.3	6000	1.5×10^7
GaN	3.5	1.5	1	1500	2.7×10^7

Wide band gap

High charge density

High electron mobility

Temperature tolerance

GaN material properties

HEMT benefits

RF System benefits

GaN offers various benefits at system level

Wide band gap

High charge density

High electron mobility

Temperature tolerance

High voltage operation

High current density

High Frequency

Higher impedance

Lower capacitance

High efficiency

High power

Smaller size

Wider bandwidth

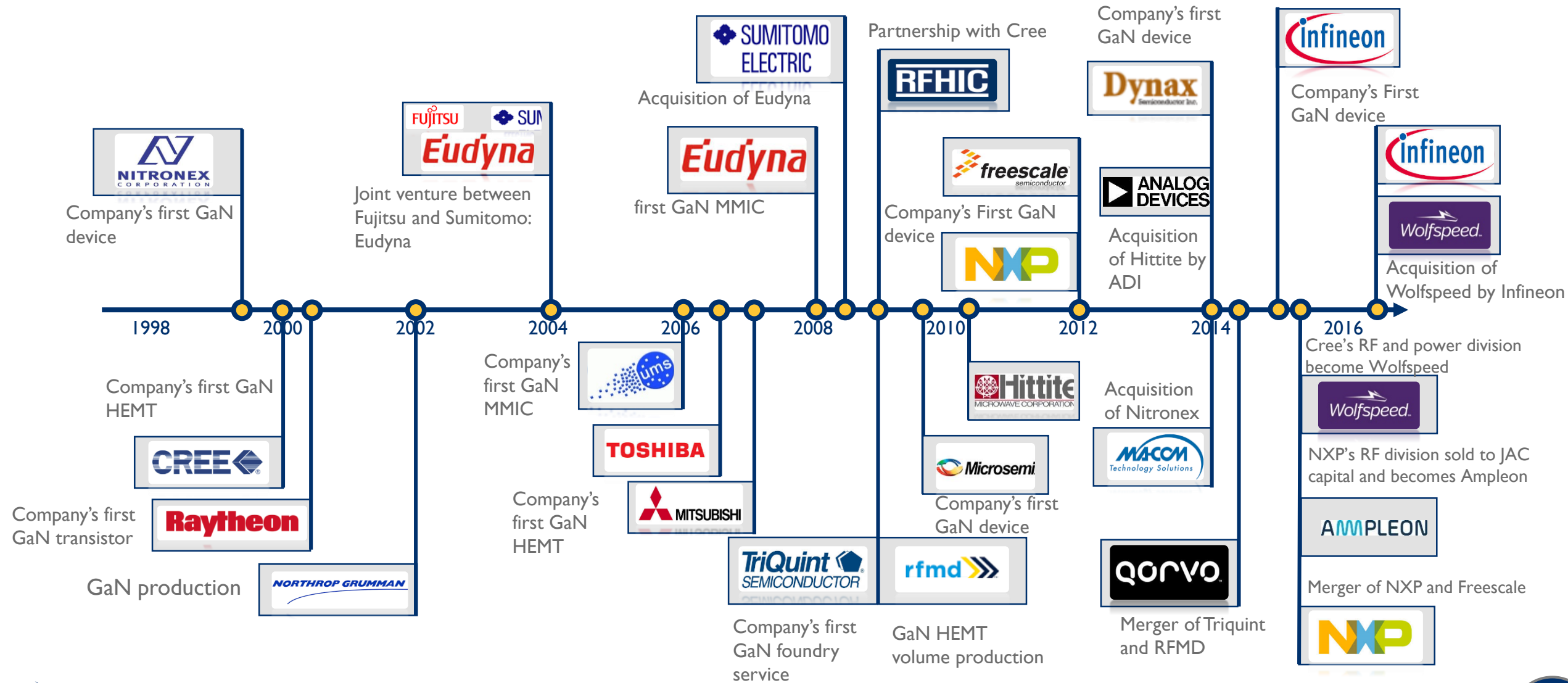
Ruggedness

Low Noise

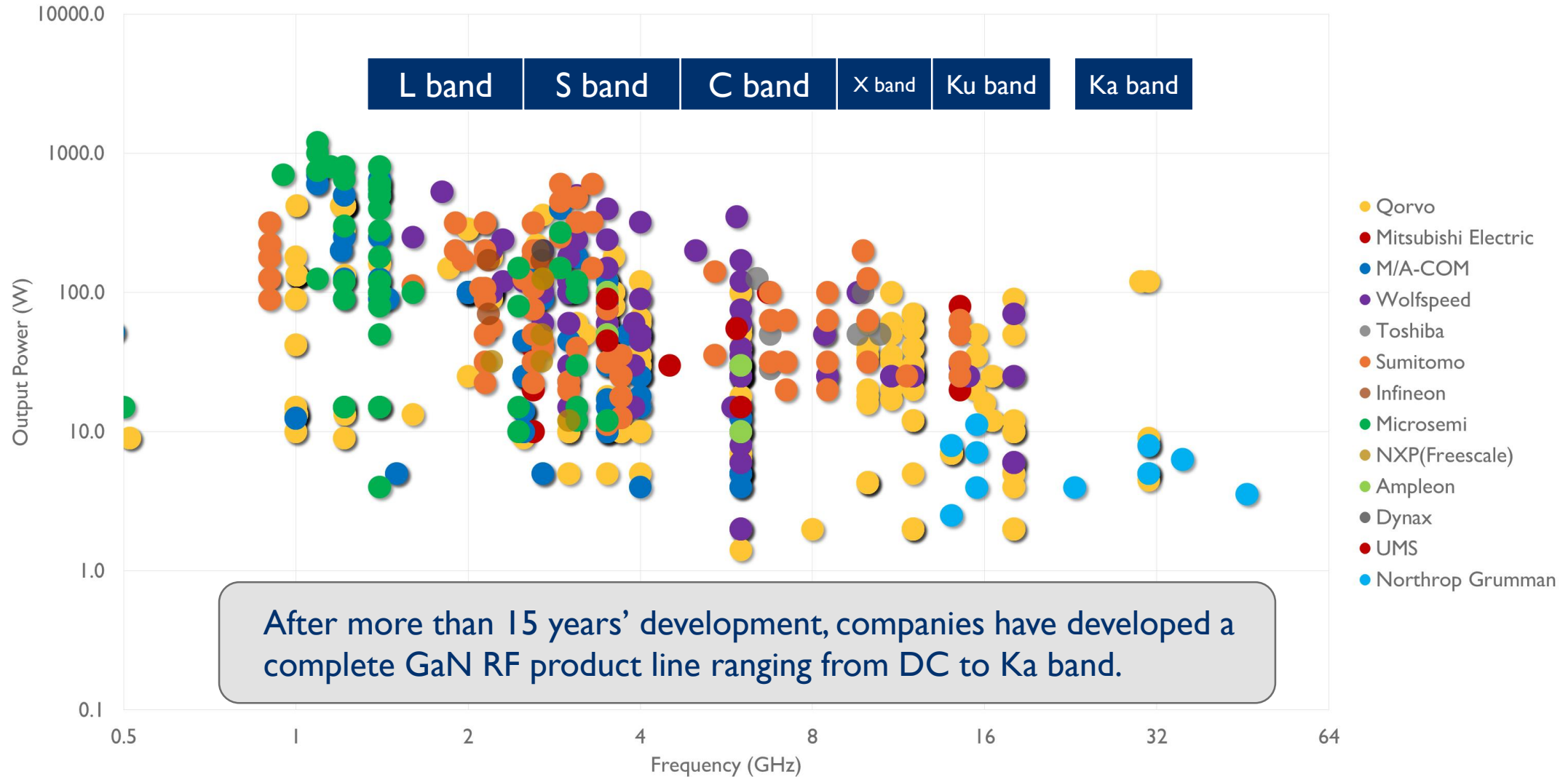
Possible applications

PA
LNA
Switch
VCO
Mixer
Gain Block
Phase shifters
Var. Attenuators

GAN RF INDUSTRY: EVENTS AND HISTORY



MAJOR GAN RF PLAYERS AND THEIR GAN PRODUCT RANGE

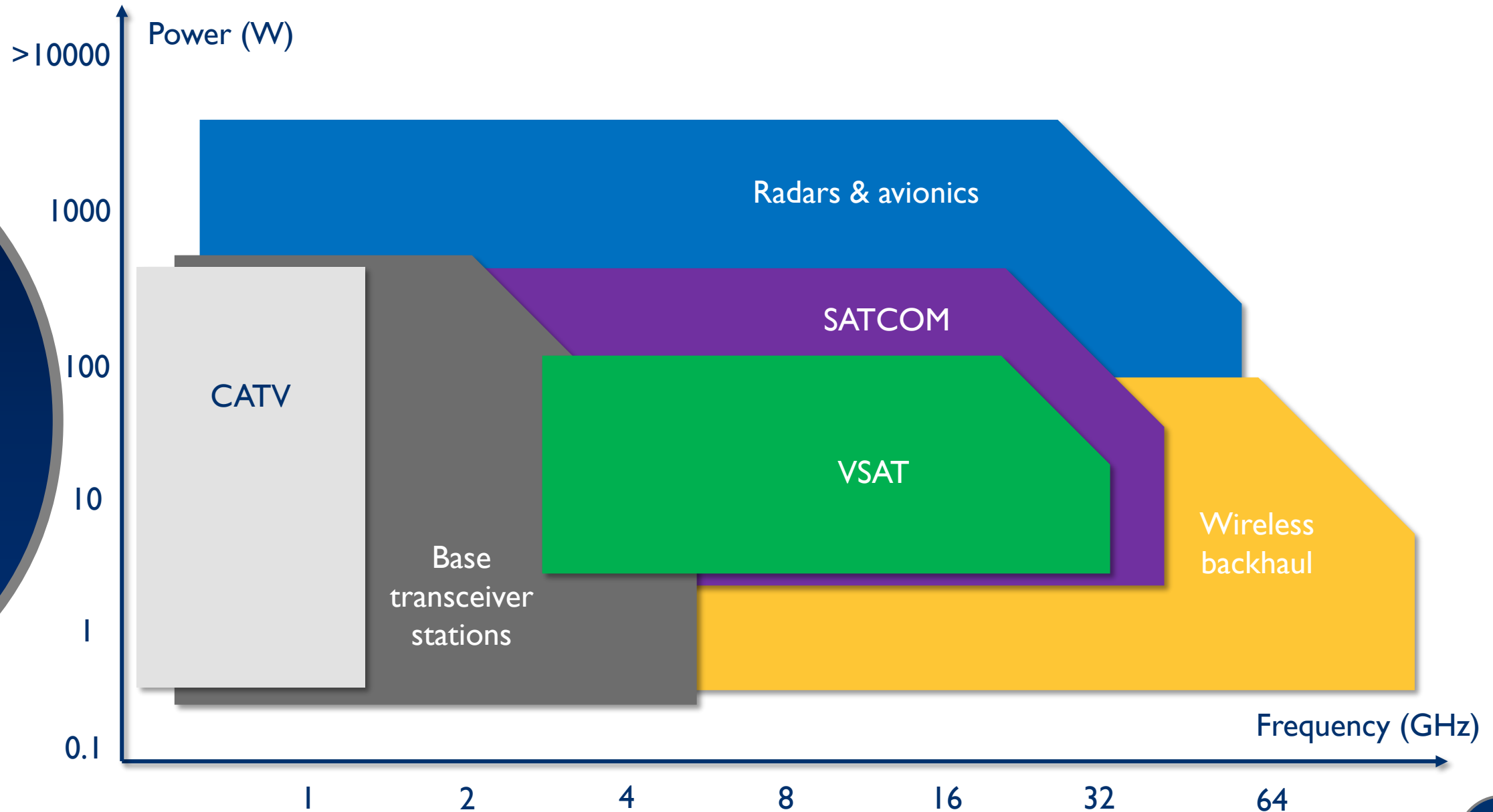


*Based on commercially available GaN product catalogues and datasheets.

WHICH APPLICATIONS ALREADY USE GAN?



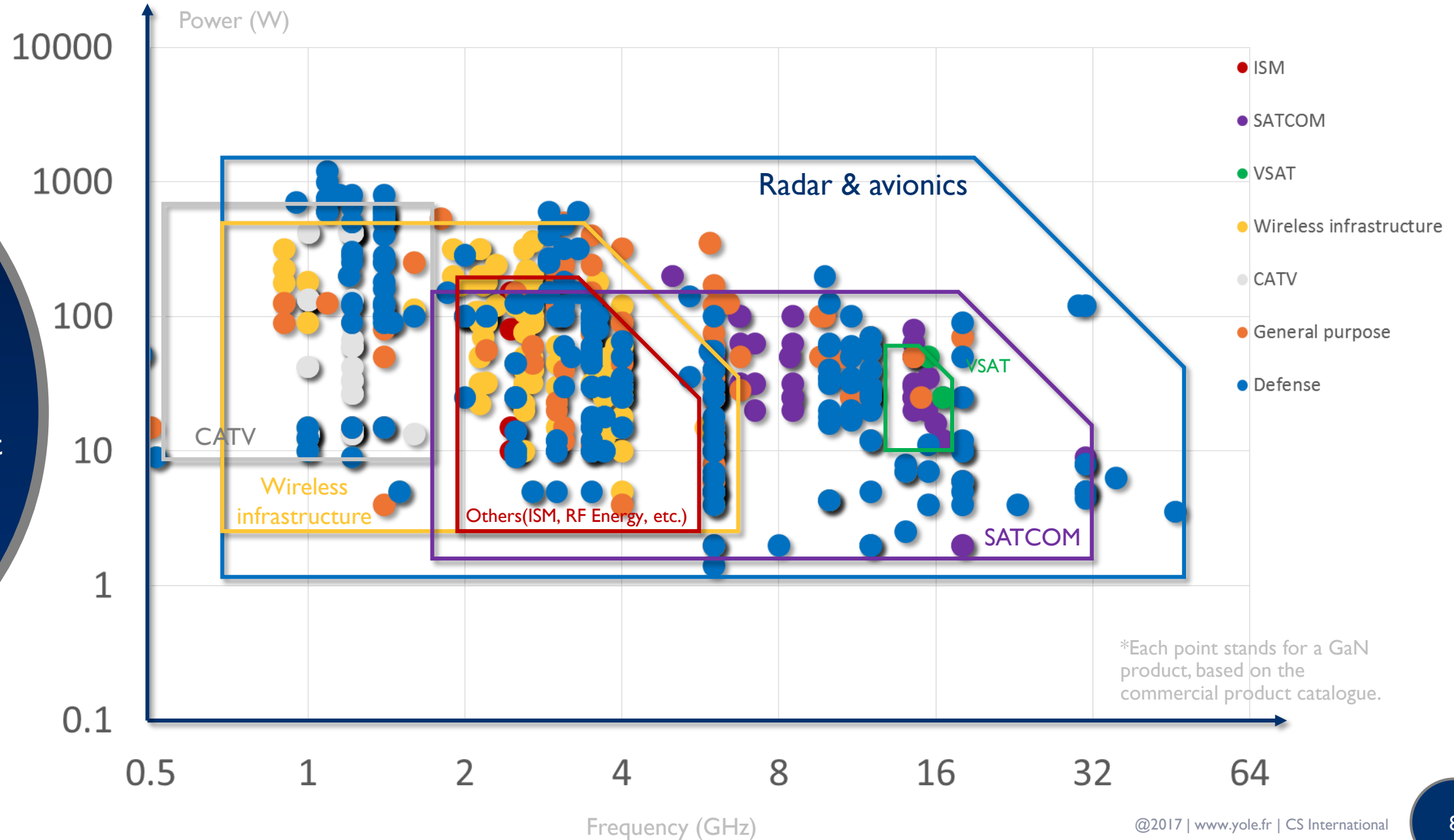
GaN's presence in the RF market is more focusing on high-power, high-frequency market



COMMERCIALLY-AVAILABLE GAN RF POWER PRODUCTS



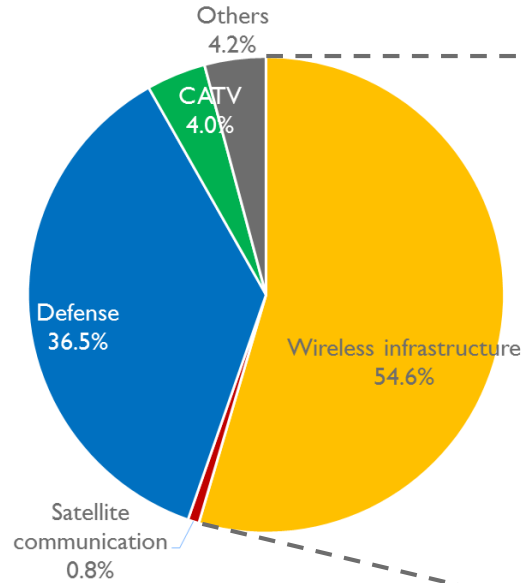
This diagram shows every commercially-available GaN product



GaN RF DEVICES: MARKET BREAKDOWN

Comparison by application: 2015 - 2022

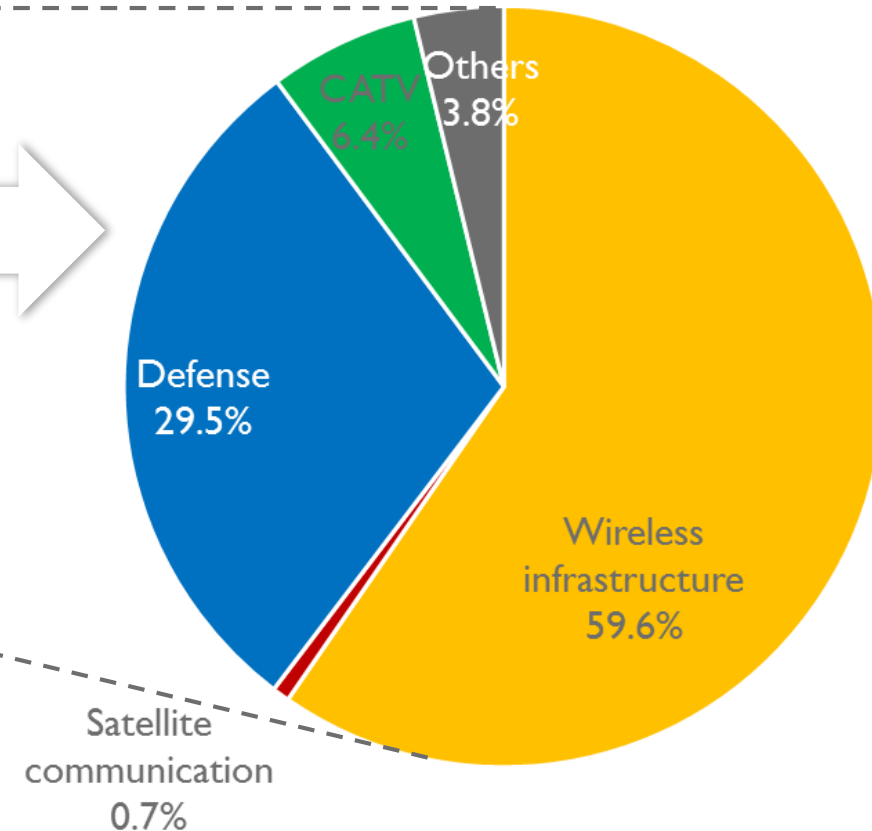
2015 GaN RF devices market, breakdown by application



TOTAL
~ \$298M

©2016
Yole Développement

2022 GaN RF devices market, breakdown by application

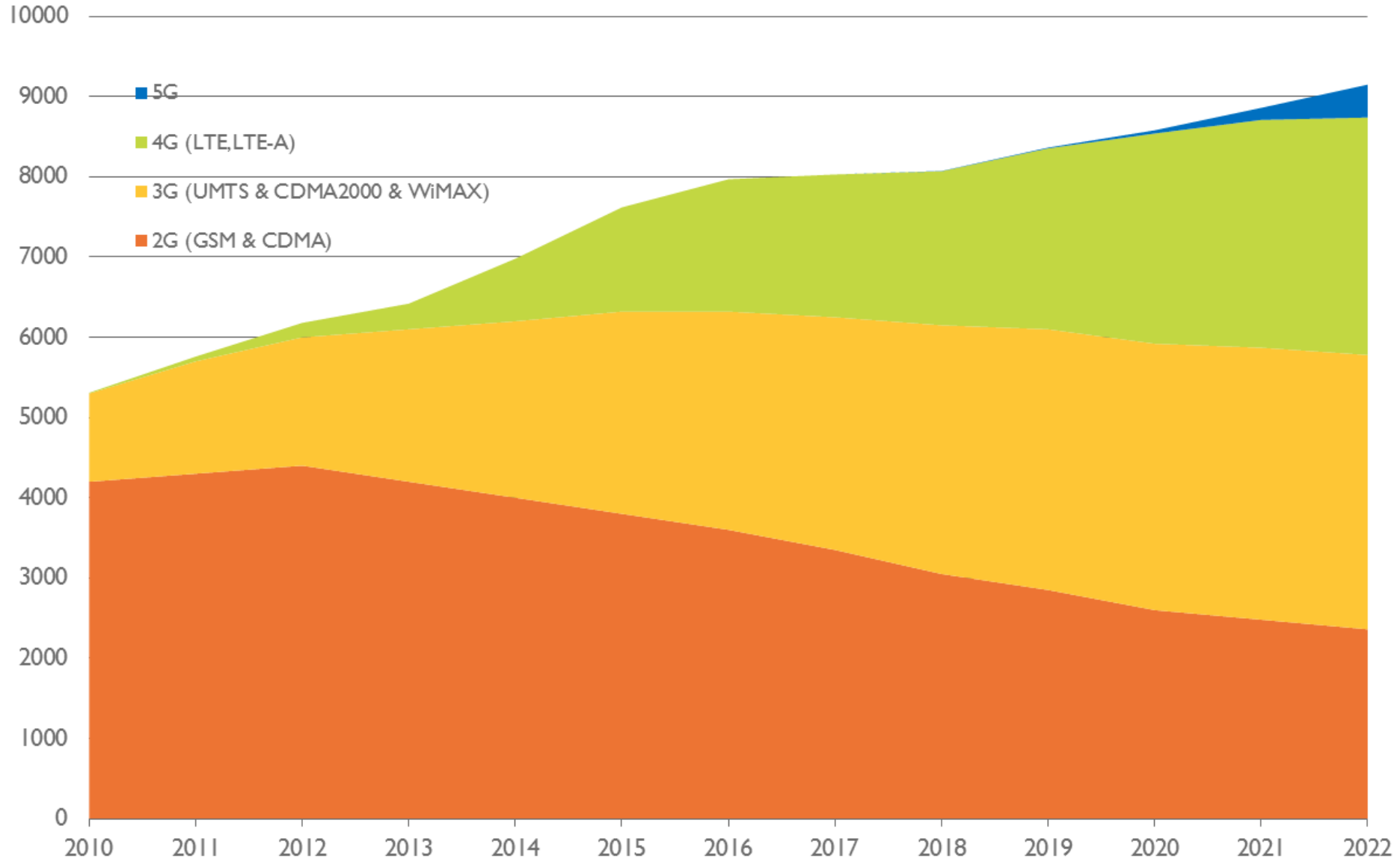


TOTAL
~ \$755M

GLOBAL CONNECTIONS BY TECHNOLOGY



- Global connections by technology (in millions, excluding M2M)



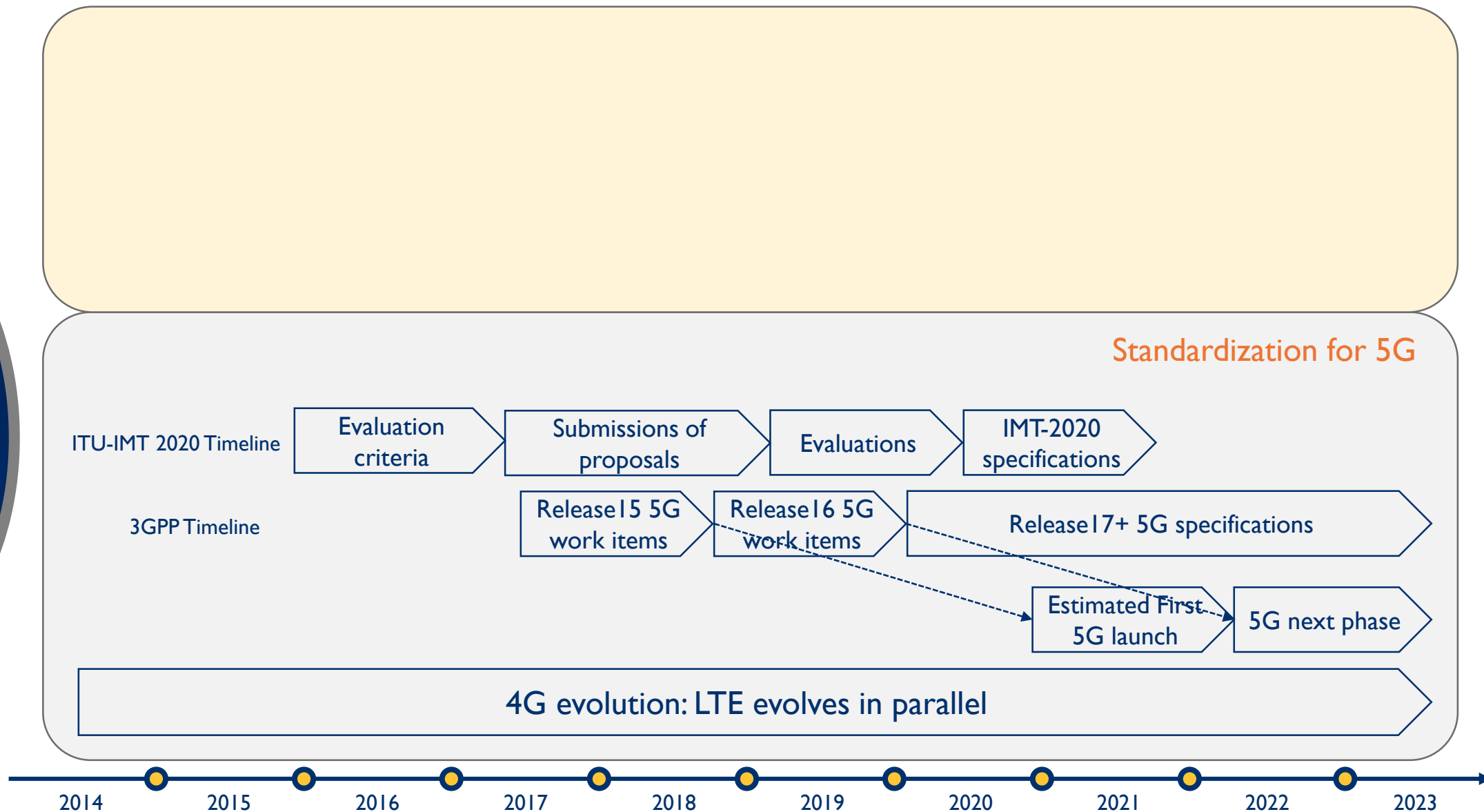
Source:
GSMA,ITU,
Yole estimation

In 2022, total connections will be 1.5x greater than 2010

GAN WILL PLAY AN IMPORTANT ROLE IN THE WIRELESS NETWORK



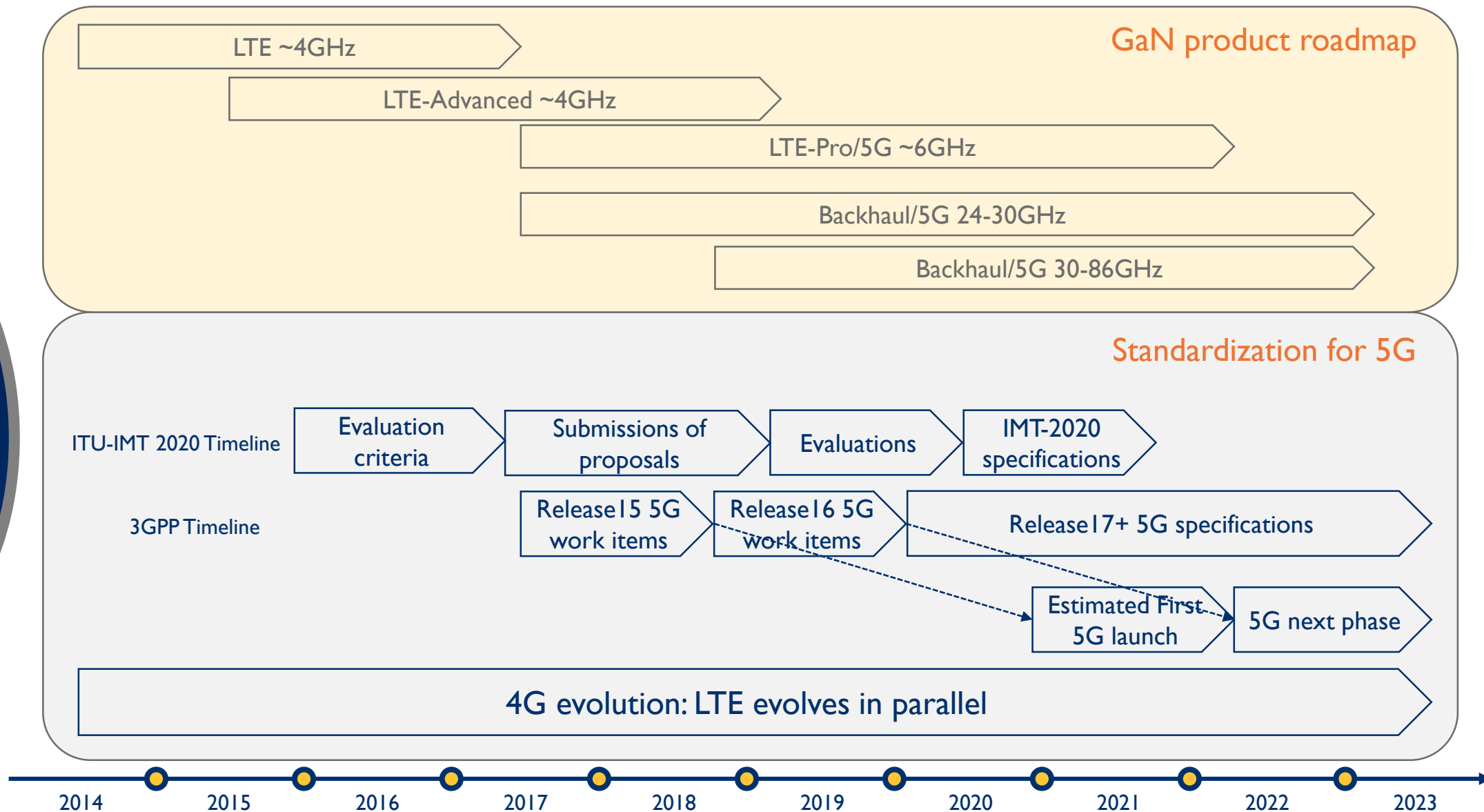
GaN will keep evolve along with the new network specifications



GaN WILL PLAY AN IMPORTANT ROLE IN THE WIRELESS NETWORK



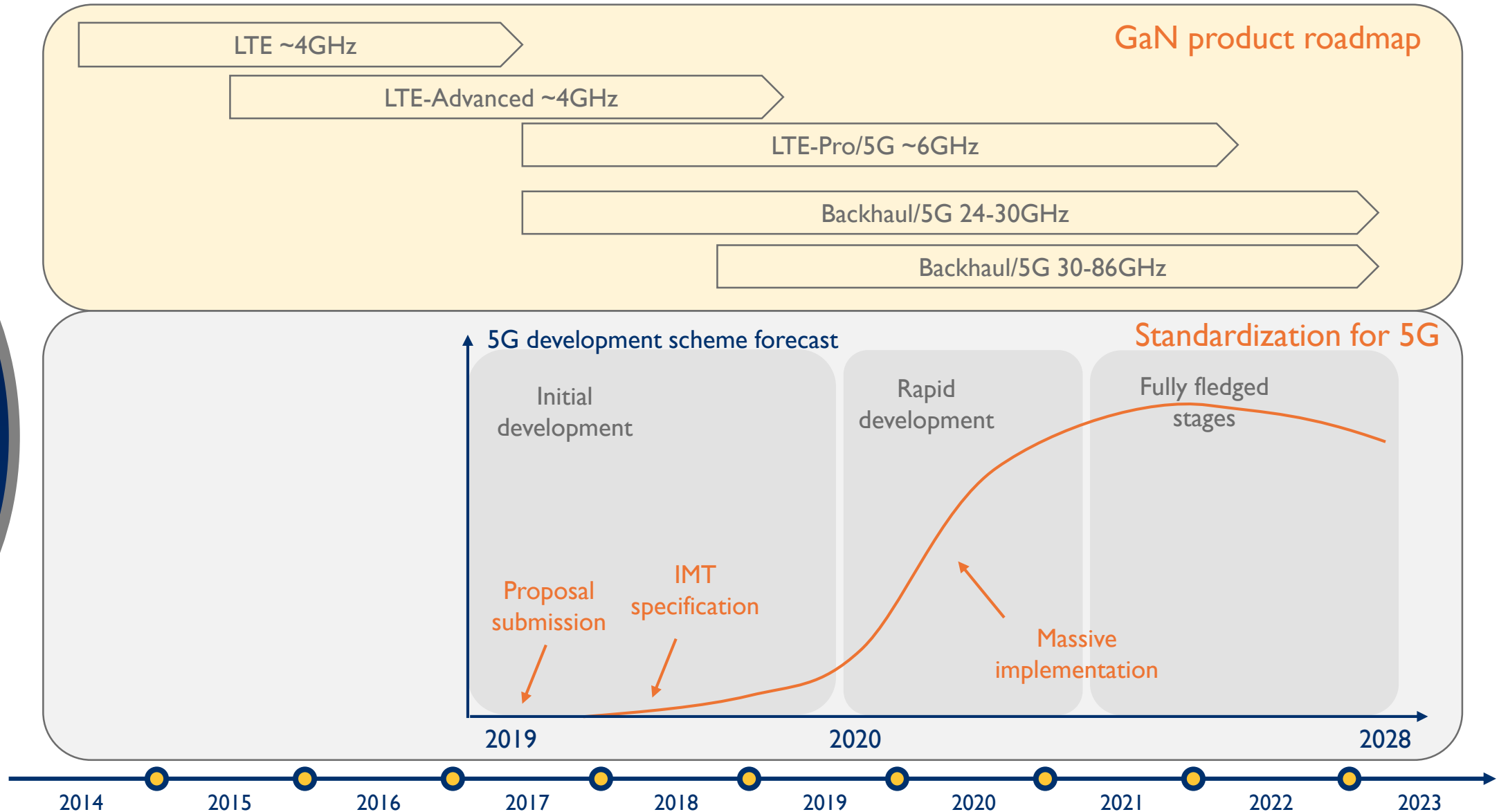
GaN will keep evolve along with the new network specifications



GAN WILL PLAY AN IMPORTANT ROLE IN THE WIRELESS NETWORK



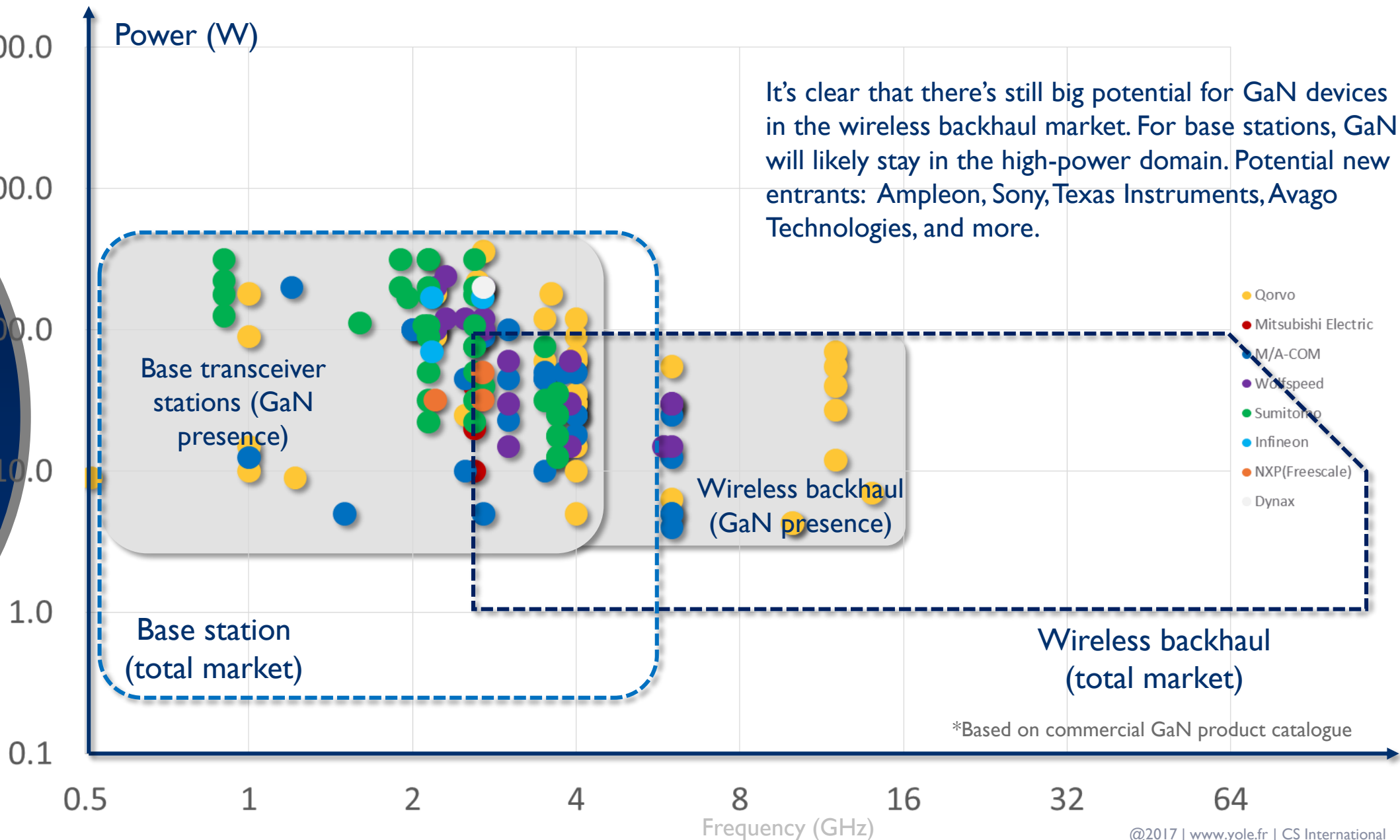
GaN will keep evolve along with the new network specifications



WIRELESS INFRASTRUCTURE: GAN RF PRODUCT SCHEME



Competition in the wireless infrastructure market is intense, since almost every GaN player has similar products



GAN-ON-SILICON VS. GAN-ON-SiC: COMPARISON



The comparison between GaN-on-Si and GaN-on-SiC has been one of the GaN RF industry's most important questions

● GaN-on-SiC pros:

- ✓ Technologically more mature
- ✓ Multi-source supplier possibility
- ✓ An established supply chain
- ✓ Better performance

● GaN-on-SiC cons:

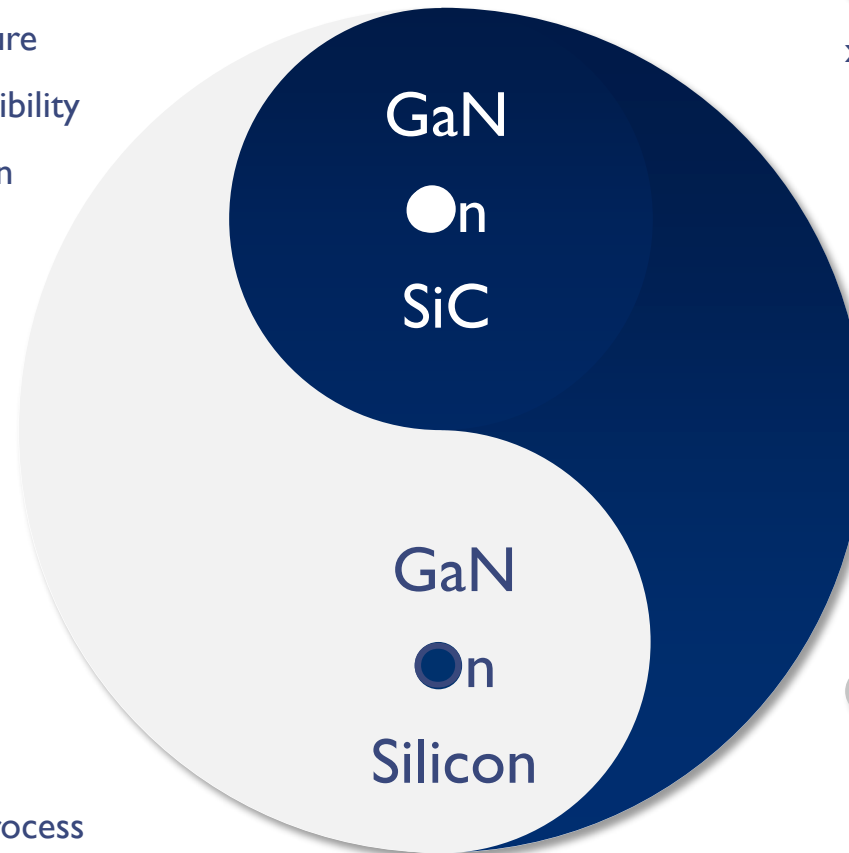
- x Relatively smaller wafer size
- x Cost, cost, cost

● GaN-on-Si pros:

- ✓ Lower cost
- ✓ Compatible with silicon process
- ✓ Sufficient (slightly inferior) performance
- ✓ Possibility of 8" (or larger) wafer production

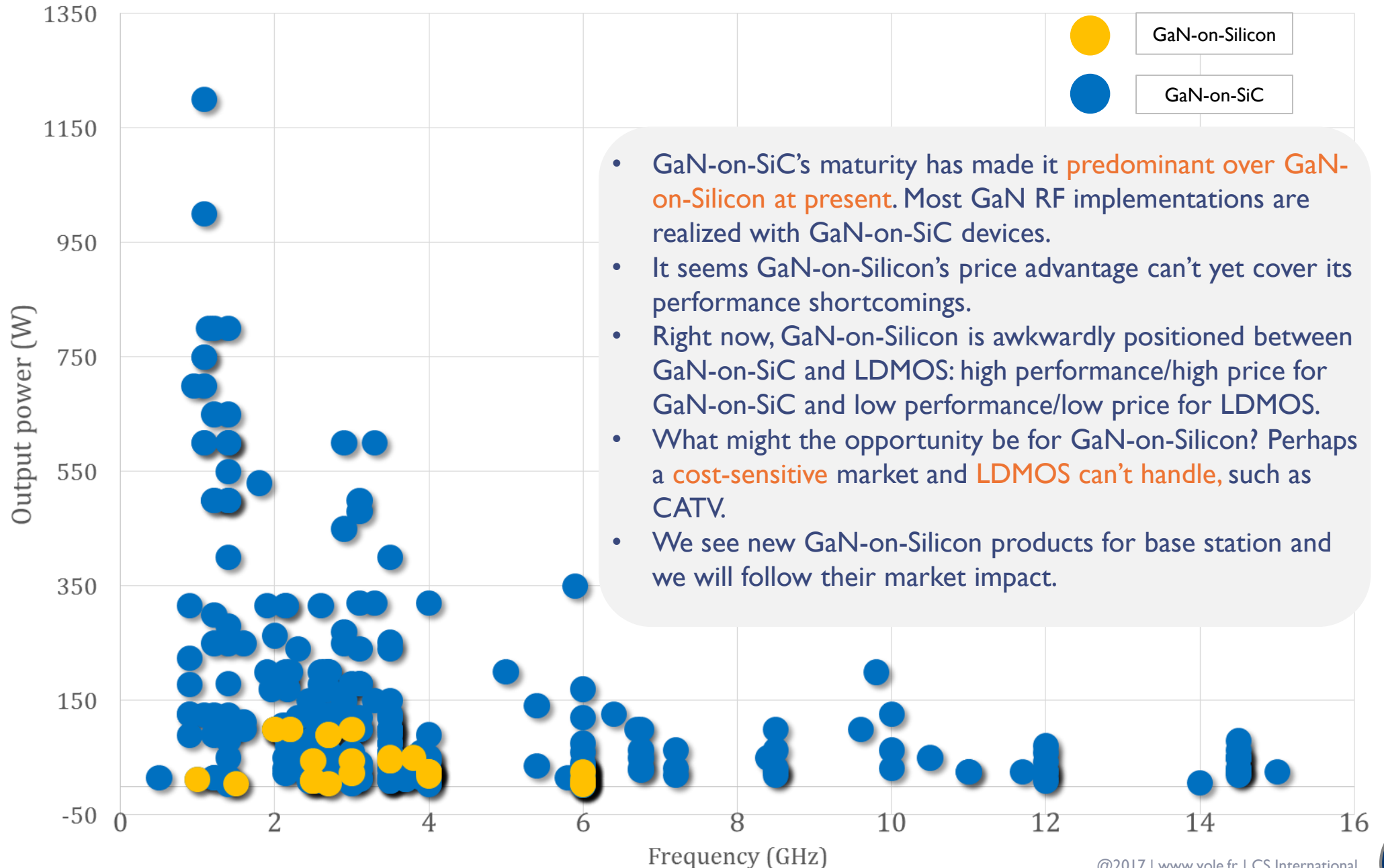
● GaN-on-Si cons:

- x Not many suppliers
- x Lower reliability level
- x Limited thermal performance
- x Still expensive compared to LDMOS



COMMERCIALLY-AVAILABLE GAN-ON-SIC-DEVICES VS. GAN-ON-SILICON DEVICES

There's no doubt that GaN-on-SiC is more popular today than GaN-on-Silicon

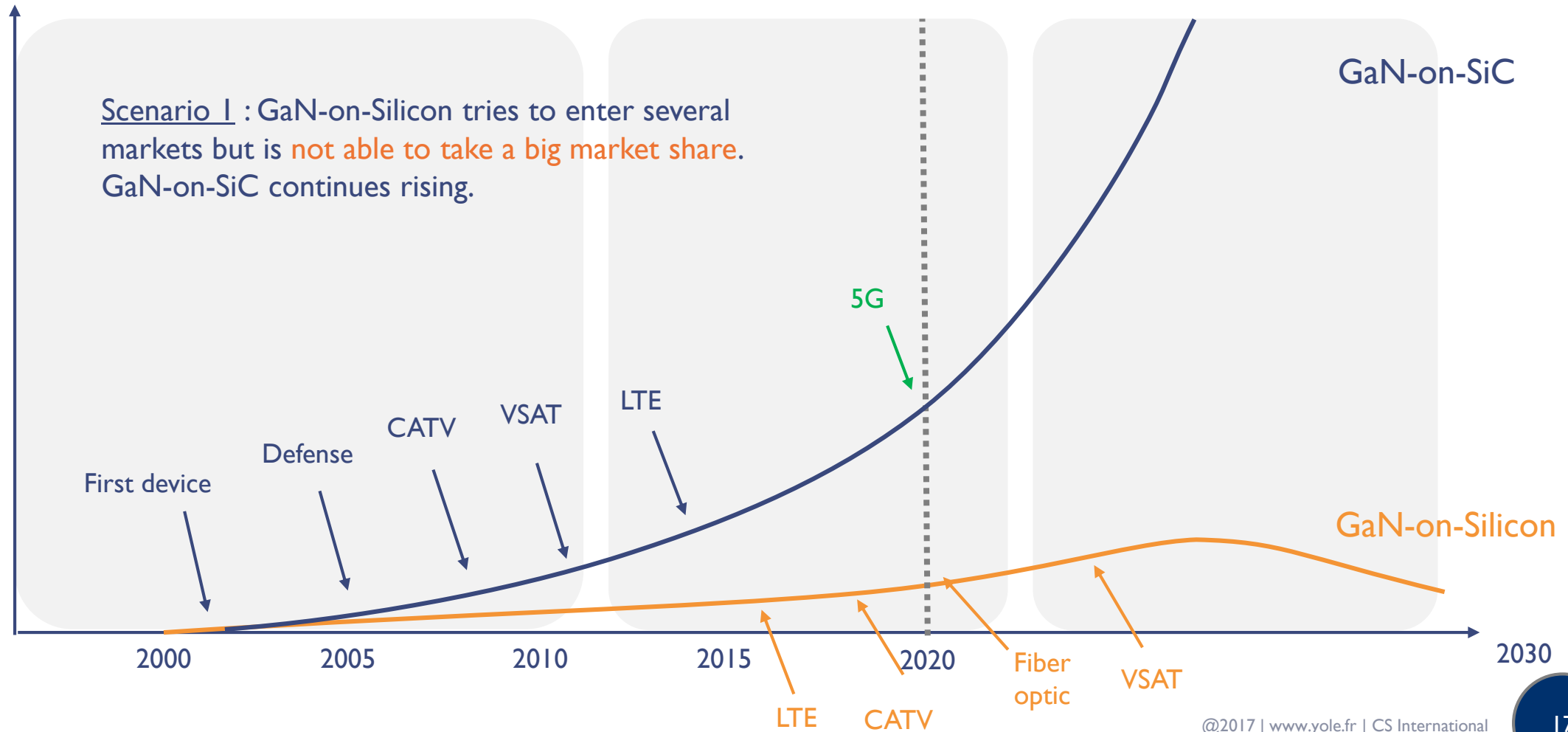


GAN-ON-SiC AND GAN-ON-SILICON: FUTURE DEVELOPMENT SCHEME

Scenario I

- Based on the current situation and available information, GaN-on-SiC will likely continue dominating the GaN RF industry **over the next five years**.

GaN-on-Silicon will become cost-driven and GaN-on-SiC will be more performance-driven

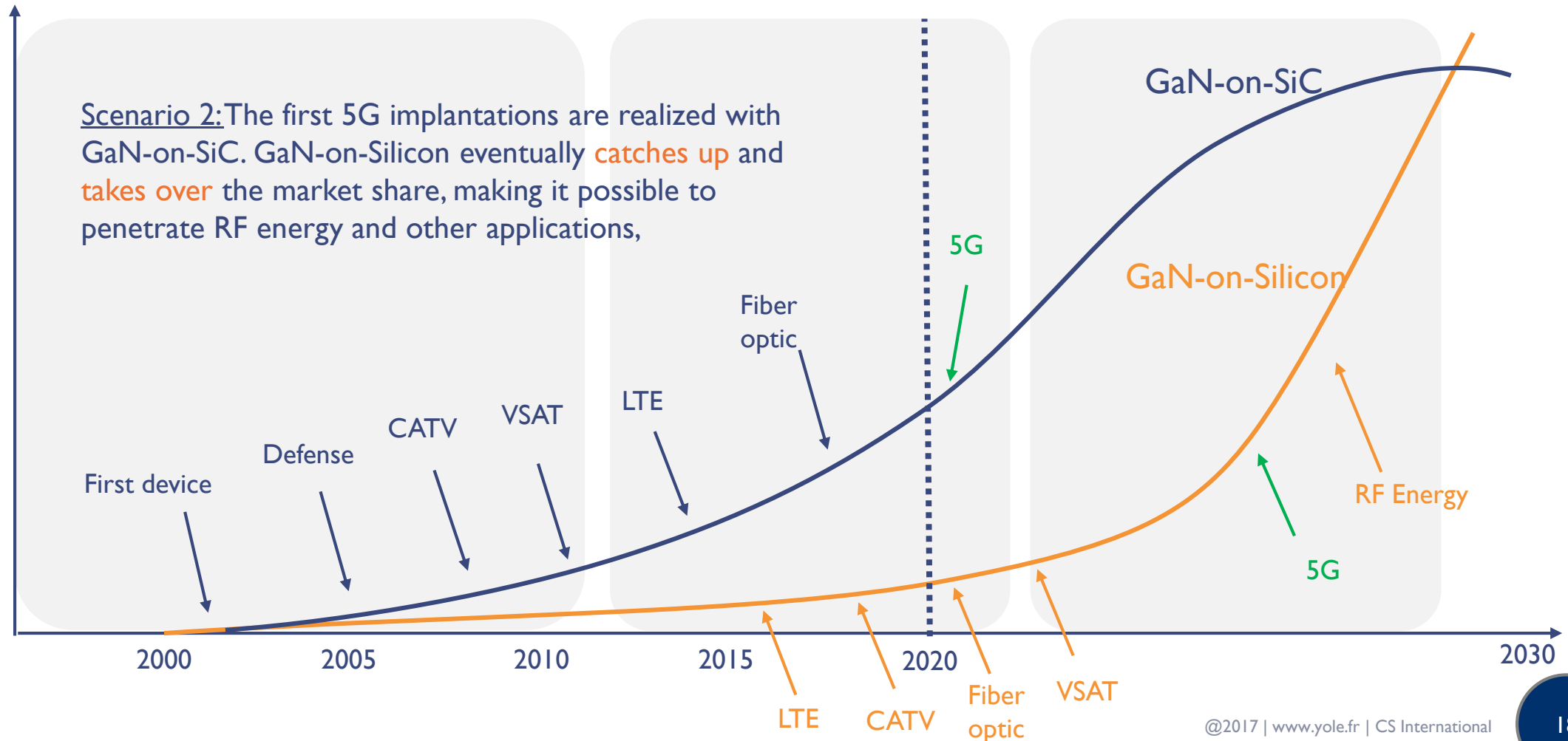


GAN-ON-SiC AND GAN-ON-SILICON: FUTURE DEVELOPMENT SCHEME

Scenario 2

- Based on the current situation and available information, GaN-on-SiC will likely continue dominating the GaN RF industry **over the next five years**.

GaN-on-Silicon will become cost-driven and GaN-on-SiC will be more performance-driven

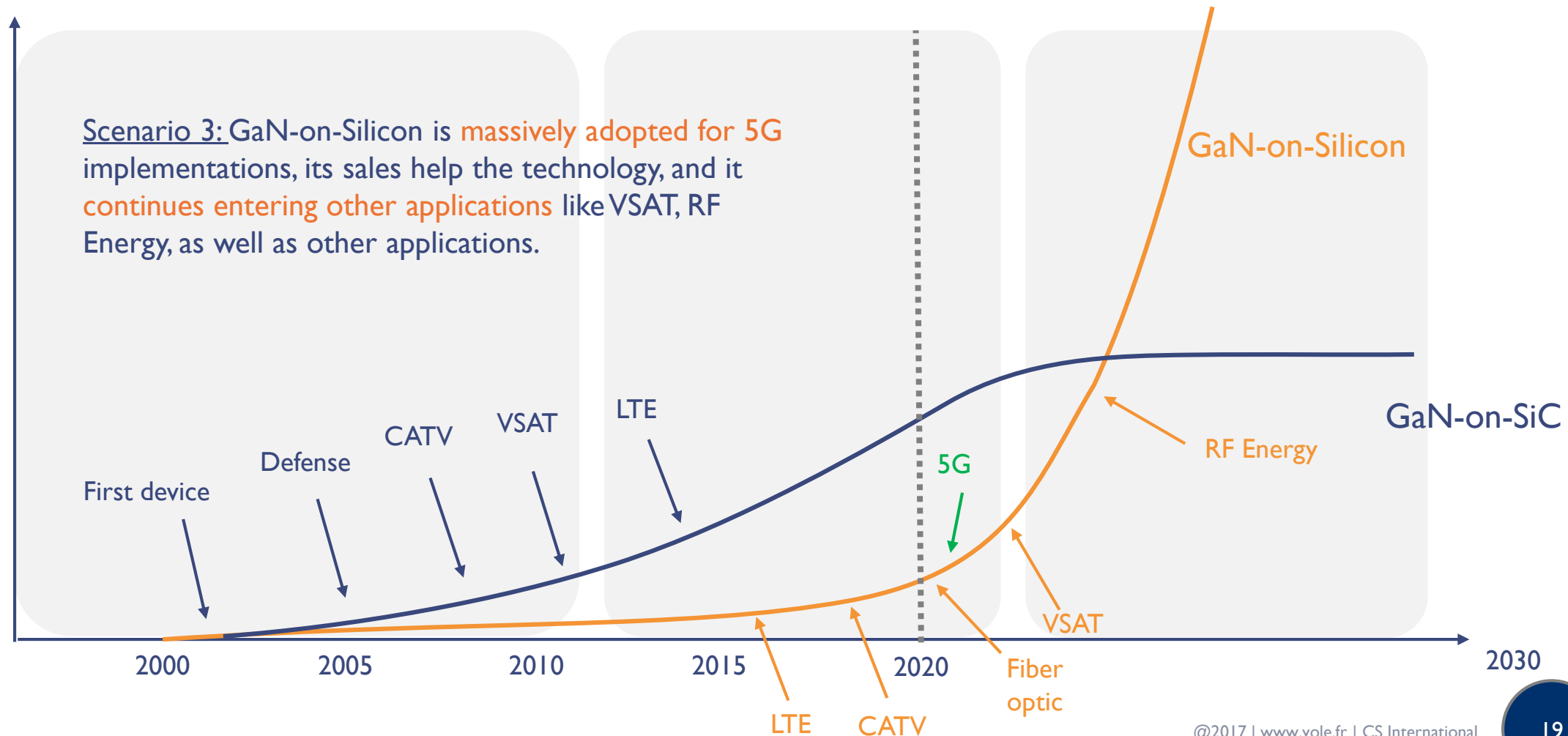


GAN-ON-SiC AND GAN-ON-SILICON: FUTURE DEVELOPMENT SCHEME

Scenario 3

- Based on the current situation and available information, GaN-on-SiC will likely continue dominating the GaN RF industry **over the next five years**.

GaN-on-Silicon will become cost-driven and GaN-on-SiC will be more performance-driven



- GaN's a prominent technology with big prospect, the industry begins embrace this new technology.
- The barrier is still the price, more effort and patience are needed.
- In the long term, GaN's future is unquestionable.